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CONTROVERSIES IN PATHOLOGY

Gleason score 3+3=6 prostatic adenocarcinoma is not benign and the current debate is unhelpful to clinicians and patients

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Summary

Prostate adenocarcinoma is a common malignancy associated with a significant morbidity and mortality. In both prostate biopsies and radical prostatectomy specimens Gleason scoring informs both treatment and outcome prediction. The current convention is that in needle biopsies, Gleason patterns 3, 4 and 5 are considered to be malignant. Despite this there is debate as to whether or not Gleason score (GS) 3+3=6 should be diagnosed as cancer due to potential over-treatment and the psychological impact on patients. It is apparent that GS 3+3=6 is indolent disease with a low risk of metastasis. However, it does have the histological features of malignancy and is capable of infiltrating the prostate gland, extraprostatic extension, and metastatic spread. Furthermore GS 3+3=6 carcinoma has immunohistochemical and molecular genetic features similar to those of higher grade prostatic carcinoma. If GS 3+3=6 tumour is considered benign, the question arises should a benign label be given to the Gleason pattern 3 component of tumour that includes Gleason patterns of higher grade? This would seem a logical step as GS 3+3=6 cancers and the pattern 3 component in cancers with multiple patterns are morphologically identical. If pattern 3 is considered to be benign, then Gleason scoring would be limited to 4+4=8, 4+5=9, 5+4=9 and 5+5=10 which is clearly inappropriate. The correct strategy to address potential over-treatment of patients with low-grade cancer is clinician and patient education, not the recalibration of Gleason grading to reclassify malignant tumours as benign.

Key words: Prostate; adenocarcinoma; Gleason score; prognosis.

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INTRODUCTION

Tumour grade is one of the most important prognostic indicators of any cancer as this is a surrogate marker of both proliferative activity and the potential to spread beyond the organ of origin. As such, tumours that are low-grade display less aggressive behaviour characteristics than high-grade cancer. By definition, all grades of cancer, including the lower grades, are considered to be malignant with architectural and cytological features of cancer and the ability to invade. In contrast, benign tumours are not able to invade but may show expansion with compression of adjacent structures.

The grading system for prostate cancer is well established and has undergone considerable evolution since the first proposals by Gleason in 1966.[1,2] It is now well recognised that Gleason pattern 1 tumours are in reality benign proliferative lesions. On needle biopsy pattern 2 tumours are not diagnosed, as well circumscribed nodules cannot be identified. Also these are rarely encountered in the peripheral zone which is the site usually sampled by needle biopsies. Gleason pattern 2 tumours may, however, be diagnosed in transurethral resections of prostate and radical prostatectomy specimens.[3] For these reasons pattern 3 cancers are considered to be the lowest grade of prostatic malignancy on needle biopsy. Beyond this, tumours with patterns 4 and 5 morphology are recognised as more aggressive cancers. Grading of prostate cancer is complicated by the fact that tumours may show a combination of these patterns. It has been clearly demonstrated that increasing amounts of pattern 4 in Gleason score (GS) 3+4=7 and GS 4+3=7 carcinomas and pattern 5 in tumours composed of two or three different grading patterns are associated with a worse outcome.[2]

In recent years, there has been a drive to label GS 3+3=6 prostatic adenocarcinoma as non-cancer. Proponents of this suggestion claim that this course of action is necessary to prevent ‘over-diagnosis and over-treatment of prostate cancer’. It is also stated that for GS 3+3=6 tumours, a cancer diagnosis is unnecessary and potentially harmful, as the patients are unlikely to suffer cancer-related morbidity and mortality. In addition, a cancer diagnosis in these cases is thought to cause anxiety and depression due to the cancer label attached to the diagnosis and that patients could be subjected to unnecessary radical therapy with associated complications

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and a potential for stress-induced self-harm.[4–6] While there has been much discussion regarding the validity of renaming GS 3+3=6 cancer as non-cancer, there are apparently difficulties in providing an alternative label for these tumours.[7–10] Suggested alternative terms include acinar neoplasm, non-cancer acinar neoplasms, prostatic neoplasms of low malignant potential (LMP), prostatic acinar intermediate neoplasms and indolent lesion of epithelial origin (IDLE).[4,11] A number of these terms are themselves confusing. If GS 3+3=6 is not cancer, then labelling it as LMP or prostatic acinar intermediate neoplasm is unhelpful, as the implication remains that these are malignant tumours. The use of the terms that label the cancer simply as a neoplasm are similarly unhelpful as they define the tumours as neither benign nor malignant. If GS 3+3=6 tumours are benign they should be called prostatic acinar adenoma. However, there can be no rational argument that this lesion is inherently benign, despite being low-risk, and importantly as pattern 3 tumour is often intimately associated with higher grade cancer. In view of this, it is somewhat surprising that this option is even considered to be a solution to issues relating to prostate cancer over-treatment, in this era of evidence based medicine that recognises the value of scientific accuracy.

MORPHOLOGY AND BEHAVIOUR

Gleason pattern 3 prostatic adenocarcinoma has morphological, immunohistochemical and molecular genetic features of cancer

Gleason pattern 3 prostatic adenocarcinoma, with or without associated higher grade cancer, has well recognised morphological features of adenocarcinoma. Architectural features of a glandular proliferation without a basal cell layer are seen in these cases. The absence of a basal layer is clear evidence of infiltration of stroma and thus these tumours exhibit invasive characteristics rather than benign accretionary growth. In comparison with benign luminal cell nuclei, the nuclei in these lesions are variably enlarged with pleomorphism and chromatin clearing, usually with prominent nucleoli, and therefore exhibit the histological features of prostatic adenocarcinoma.[1,2,11] These tumours also display overlapping immunohistochemical findings and molecular genetic alterations with higher grade cancer.[12–15] These include over-expression of alpha-methyl-acyl-coA racemase, hypermethylation and down-regulation of glutathione S-transferase, EGFR gene mutations and TMPRSS2–ERG gene fusions.[13,14]

Gleason pattern 3 cancer has the ability to invade host tissue

Prostate carcinoma spreads by infiltration of prostatic stroma, infiltration of peri-neural spaces and by extension of the tumour beyond the prostatic pseudocapsule. GS 3+3=6 tumours clearly showing infiltration of prostatic stroma and perineural invasion are well recorded (Fig. 1).[16] Extraprostatic extension (EPE) is also an established feature, being reported in 0.28% of cases in one series[17] and 6.3% in another series of GS 3+3=6 prostatic adenocarcinomas.[18] In this latter series EPE was described as focal in 3.9% of cases, while 2.4% had non-focal or established EPE. Additionally, in the latter study, seminal vesicle involvement was found in one of 3,288 cases of GS 3+3=6 prostatic adenocarcinoma treated by radical prostatectomy. It is clear from these data, that while infiltration of GS 3+3=6 cancers beyond the prostate is relatively rare, the tumours do have the ability to infiltrate adjacent tissues. An observation that appears to have been neglected by those who claim that GS 3+3=6 is not cancer, is when there is Gleason pattern 3 tumour in combination with a higher grade component, it is not uncommon for extraprostatic tumour or a metastasis to have a pattern 3 component.

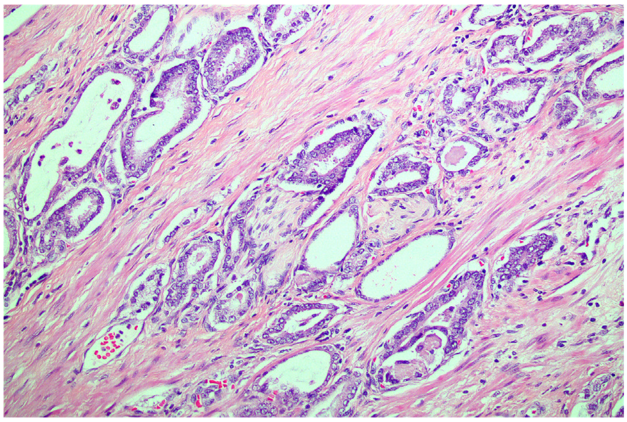

Fig. 1 Gleason pattern 3 prostatic adenocarcinoma infiltrating prostatic stroma with perineural invasion.

GS 3+3 = 6 adenocarcinoma is not incapable of metastasising and spreading outside the prostate

It has been stated that pure GS 3+3=6 adenocarcinoma is incapable of metastasising and spreading outside the prostate[4] and several studies have been cited as evidence of this claim.[19–21] Gleason pattern 3 cancer may also occasionally be seen infiltrating extraprostatic fat (Fig. 2), seminal vesicle (Fig. 3) and lymph nodes (Fig. 4) which is clearly not a feature of benign disease.

In a study by Donin *et al.* of 857 men with GS 3+3=6 cancer treated with radical prostatectomy, 2.7% of patients developed biochemical recurrence.[19] This was interpreted as local recurrence rather than metastatic spread, as there was a prostate specific antigen decline after radiotherapy. These results were somewhat surprisingly taken as evidence of incapability of GS 3+3=6 cancer to metastasise. Follow-up of this series ranged from 3 months to 11 years, with a median follow-up of 6 years. It is noteworthy that these tumours were

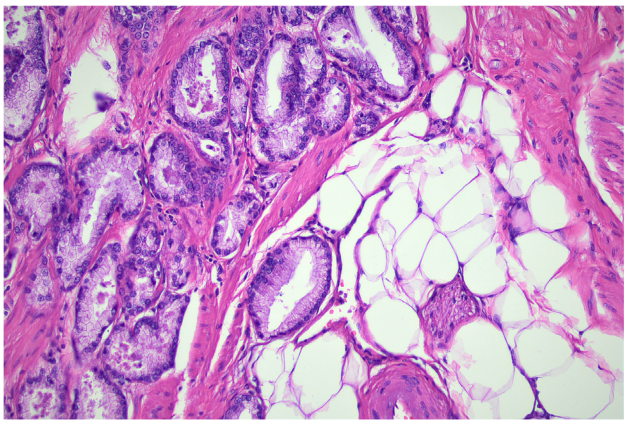

Fig. 2 Infiltration of Gleason pattern 3 prostatic adenocarcinoma into periprostatic fat.

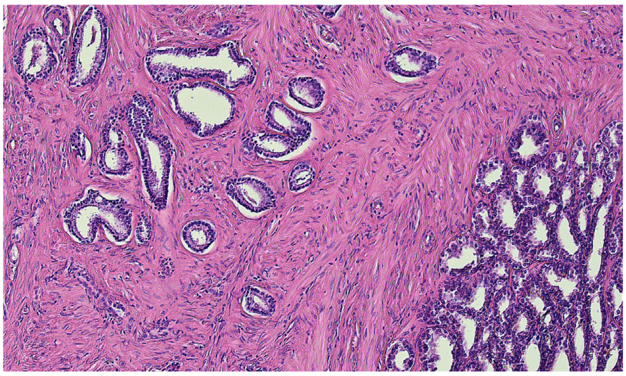

Fig. 3 Infiltration of Gleason pattern 3 prostatic adenocarcinoma into seminal vesicle.

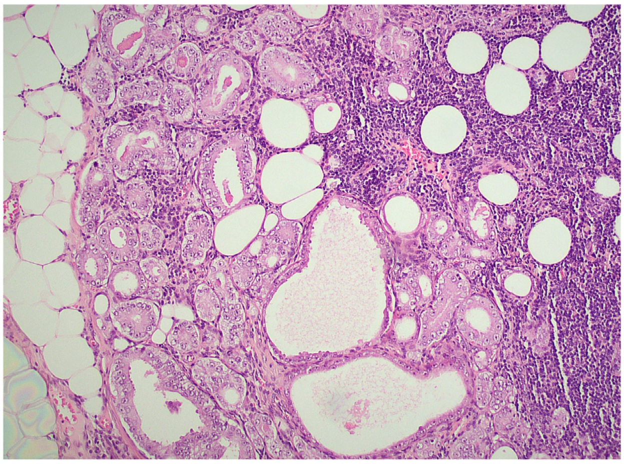

Fig. 4 Lymph node with infiltration of Gleason pattern 3 prostatic adenocarcinoma.

removed at radical prostatectomy, with follow-up radiotherapy as appropriate. Clearly these findings cannot be used to define the natural history of the disease as the tumour has been removed surgically, with any recurrent tumour ablated by radiotherapy, thus preventing future metastatic events or denying the tumour the opportunity to evolve into a more aggressive phenotype. In a study by Ross *et al.*,[20] cases were reviewed from a large series of tumours that were reported as GS 3+3=6 adenocarcinoma, with 19 having lymph node metastasis. These tumours were all found to be of higher grade leading to the conclusion that GS 3+3=6 prostatic adenocarcinoma has no potential to metastasise. Absence of GS 3+3=6 cancers in a series is not evidence that these tumours have no ability to metastasise, leaving the validity of the conclusions based on this study seriously open to question. In a study of 1,101 cases of prostate cancer reported by Kweldam *et al.*, none had metastasised and none of the 3.8% of patients who died of prostate cancer had a GS 3+3=6 at radical prostectomy.[21] Again, absence of progression in a series of surgically resected low-grade tumours such as these cannot be taken as evidence of incapability of the tumour to metastasise. What it does indicate, however, is that the tumours are of low risk, which does not amount to universally benign behaviour. A study from Australasia focused on 496 patients who were considered unsuitable for radical prostatectomy.[22] This series included 19 cases of GS 3+3=6 cancer. All cases were subject to central pathology review and after 10 years follow-up, six of the patients had died of disease. In a SEER database study of 21,960 patients, the prevalence of nodal metastasis of GS 6 cases was 0.48% not 0%.[23] In another study, the 15 year prostate cancer specific mortality for GS $\leq$6 cancer was 0.2–1.2%.[24] The prospective ProtecT randomised trial investigated outcome in 1,610 men managed with active monitoring, prostatectomy and/or radiotherapy, with a follow-up of 15 years. Metastases subsequently developed in 104 cases and, of these, 53 had GS 3+3=6 tumours in the baseline biopsy, while 49 of these patients were identified as having low-risk disease according to the Cancer of the Prostate Risk Assessment (CAPRA) criteria. This trial highlights the biological aggressiveness of these tumours despite the initial, apparently low-risk, biopsy histology.[25]

While it might be appropriate to question the validity of assigned grades in the SEER study due to the absence of central review, collectively these results indicate that GS 3+3=6 prostate cancer is not benign, is not incapable of direct spread beyond the prostate or metastasising, and that the tumour has a low risk of these events. The significance of association of Gleason pattern 3 cancer with higher grade cancer[1,2] also argues against tumours of this morphology being labelled benign.

Criteria for diagnosing cancer should be consistent and not dependent on accompanying higher grade cancer

Those advocating renaming of GS 3+3=6 cases as benign have not discussed critical practical aspects relating to this recommendation. If Gleason pattern 3 is considered benign, does this apply to tumour in its pure form only or does it include tumour associated with higher grade cancer?[4–6] It can be argued that there should be consistency in diagnosing pattern 3 cancer whether or not it is accompanied by higher grade disease. To suggest that pattern 3 on its own is benign, but when identical tumour is accompanied by higher grade cancer it is considered malignant, is both inappropriate and scientifically unsound.

Another issue relates to tumours with Gleason pattern 2 in transurethral resection of prostate and radical prostatectomy specimens. If Gleason pattern 3 is labelled benign does this also apply to Gleason pattern 2? This has been neither validated nor even considered.

If pattern 3 cancer is considered to be benign, even when associated with higher grade cancer, this raises a number of practical reporting issues and this would not be an uncommon problem. This is not simply of academic interest as Gleason pattern 3 tumour is often intimately admixed with one or more higher Gleason grades. As is frequently encountered, different cores in a biopsy series may have differing Gleason patterns and scores. As a consequence, samples of the same tumour nodule may be reported as benign and malignant.

To label cores with GS 3+3=6 tumour as benign and other cores with higher grade cancer as malignant would promote confusion relating to estimation of the potential behaviour of the tumour, particularly in terms of the volume of cancer present. This would be particularly relevant if there were multiple cores with large volumes of GS 3+3=6 cancer in a core biopsy along with occasional cores containing higher grade cancer. If the GS 3+3=6 tumour is considered benign then there will be an underestimation of the volume of cancer present in the entire case.

In mixed grade nodules in radical prostatectomies, if the grade 3 component is treated as non-cancer, then assessment of volume should be only of the high-grade component. The current available volume estimation methods would be impossible to apply in these cases.[26] Even if a method is devised to measure the volume of only the high-grade component, the result would not be of value in accurately predicting outcome as there would be an underestimation of the volume due to exclusion of the pattern 3 component.

Gleason pattern 3 cancer in a biopsy may be associated with unsampled higher grade cancer. For this reason, upgrading following radical prostatectomy occurs in a proportion of these cases.[27,28] While the use of multiparametric magnetic resonance imaging (mpMRI) scans and targeted biopsies have reduced this, there is still an approximate actuarial risk of upgrading in 20% of cases.[29] While upgrading of cancer is understandable because of recognised sampling problems with needle biopsies, it would be difficult to explain how a lesion that was initially reported as benign was subsequently labelled cancer on follow-up biopsy. The risk of progression to a higher grade also cannot be excluded, for as noted earlier, outcome information specifically relating to cancer deaths based on radical prostatectomy findings is not a true indication of the risk as the tumour has been resected and its natural history aborted.

The Gleason grading system relates to malignant tumours and as a consequence it would be inappropriate to apply GS 3+3=6 grading to lesions described as non-cancer acinar neoplasms.[1,5] If GS 3+3=6 prostatic adenocarcinoma is considered not to be cancer, then a GS should not be applied and Gleason pattern 3 tumour should not be included in higher GSs. This would mean that the Gleason grading system would consist of patterns 4 and 5 only and GS 3+4=7 and 4+3=7 would both be considered GS 4+4=8 instead of GS 7. In this scenario, the whole of Gleason system would be relegated to four categories of GS being 4+4=8, 4+5=9, 5+4=9 and 5+5=10. This is clearly unacceptable as the current GSs and International Society of Urological Pathology (ISUP) grades have been shown to be of prognostic significance, with low grades indicating lower risk and increasing grades correlating with worse outcomes.[30] The current valuable information provided in GS 7 cases by separating GS 3+4 from GS 4+3 cases,[31] and providing the volume of pattern 4 as a percentage,[32] would also no longer be available to influence management, as only the pattern 4 component would be considered to be cancer.

CLINICAL ISSUES

Reporting GS 3+3 = 6 cancer when there are other indicators of possible aggressive disease

If radiological findings indicate significant cancer, for example if mpMRI scans show a $\geq$3 Prostate Imaging and Reporting Data System (PI-RADS) score lesion and/or a prostate-specific membrane antigen (PSMA) positron emission tomography/computed tomography (PSMA-PET/CT) scans show an avid lesion, then labelling a GS 3+3=6 cancer as benign would cause confusion with regards to management. Should such a patient have active surveillance or definitive radical treatment? If definitive radical treatment is deemed necessary, it would be difficult to explain to the patient why he should have such treatment despite having benign disease.

How to manage GS 3+3=6 prostate cancer in the presence of intraductal carcinoma also becomes even more problematic than it currently is if the invasive component is considered benign. Currently, the ISUP recommends inclusion of the grade of intraductal cancer in the GS if there is any invasive cancer, which means the clinician and patient have the appropriate information to receive the necessary correct treatment.[33] If the invasive component is GS 3+3=6 and is labelled benign, the patient has a high risk of not receiving appropriate treatment, as the recommendation is not to include the intraductal component in the GS if there is no invasive cancer. Using the criteria of the North American Genitourinary Pathology Society,[34] which does not allow inclusion of the intraductal carcinoma grade in the GS, whether or not there is stromal invasive cancer, would mean that a benign label for the GS 3+3=6 component would prevent appropriate treatment in any event.

GS 3+3 = 6 cancer on needle biopsy and active surveillance

Treating GS 3+3=6 as cancer would mean that clinicians and patients are appropriately informed regarding the low-grade and low-risk nature of this grade of cancer and would encourage active surveillance if indicated, bearing in mind that not all 3+3=6 cancers remain organ-confined in the long term. A variety of active surveillance protocols are available, some of which have been endorsed internationally, and these provide guidance regarding patient selection and long-term management.[35]

In some active surveillance protocols, inclusion criteria allow only GS 3+3=6 cases, while others permit a small volume of Gleason pattern 4.[35] Some of these protocols include the volume of cancer, with $\leq$50% involvement of any core necessary for a patient to be enrolled in active surveillance, and mislabelling GS 3+3=6 cancer as benign may prevent the patient qualifying for active surveillance, which would appear unnecessary for a 'benign' lesion. Even if the patient is recommended for active surveillance, there may be less inclination by the patient to adhere to the recommendations and onerous follow-up if there is no cancer diagnosis. This may lead to potential delays in diagnosing an underlying high-grade cancer. The problem would be further compounded if the patient was subsequently shown to have >50% GS 3+3=6 tumour in a core as this could lead to the patient failing to receive definitive and potentially curative treatment should it be labelled benign.

GS 3+3 = 6 prostate cancer should not be re-labelled for the purpose of preventing over-diagnosis and over-treatment of prostate cancer

It is the role of the pathologist to provide an accurate diagnosis for surgical specimens and thus inform the appropriate treatment by surgeons and oncologists. As such it is inappropriate for pathologists to best guess a diagnosis on the basis of perceived behaviour of the tumour, while neglecting the morphology. It is not surprising that 278 of 314 (89%) of pathologists surveyed by ISUP endorsed retaining Gleason pattern 3 lesions as carcinoma.[7]

Gleason pattern 3 carcinoma has the histological features of malignancy and to report it otherwise would provide unhelpful information to the referring clinician. The management of prostate cancer has evolved considerably over the past two decades and beyond treatments resulting in the extirpation of the prostate by surgery, cryotherapy or microwave ablation, or by radiotherapy for cases where surgery is inappropriate, monitoring of the patient by active surveillance is an increasingly viable alternative. Whatever opinion the patients may already have about their treatment, it is up to the clinician to provide all the information available regarding a diagnosis of Gleason pattern 3 cancer and, in particular, explain that not all cancers are equal and that this is a low-risk disease. There are many low-grade, low-risk cancers in other organs, particularly in the skin, which patients accept as non-life-threatening, even with a cancer label. Importantly, for the reasons outlined above, a diagnosis of GS 3+3=6 prostate adenocarcinoma is not an over-diagnosis, it is the correct diagnosis.

In some instances, the issue is one of over-treatment and here clinician education would be the most appropriate course of action. This is far more acceptable from a scientific viewpoint than changing a cancer diagnosis to benign simply to encourage the abandonment of erroneous practice by clinicians. Interestingly, over-treatment has been found far more commonly in some institutions than others.[19] There also seems to be considerable variation in the number of cases of Gleason 3+3=6 reported from different institutions. In the study by Donin *et al.*, GS 3+3=6 cancer was found in >48% of radical prostatectomy cases.[19] In contrast, GS 3+3=6 cancer was found in only 7.5% of a series of 2,900 cases from Australia,[30] with clinically so-called insignificant cancer (defined as organ confined, with a volume of <0.2 cc and a GS of <7)[36] being found in only 5.2% of cases. This raises questions as to the comparative grading ability of pathologists. It is well recognised that there is only moderate reproducibility of prostate cancer grading amongst pathologists and upgrading of GS in tumours diagnosed as GS 3+3=6 has been reported.[20,37,38] It is our experience with the ISUP Imagebase protocol that GS 3+3=6 tumours may be assigned a GS of 3+4=7, while higher grade tumours may be graded from 3+3=6 to 4+4=8, even by experienced uropathologists.[39] It is for this reason that cases in outcome studies should undergo central review by experts rather than relying on the GS provided in the histology report. Difficulties relating to inconsistency of prostate cancer grading amongst pathologists could also seriously impact patient treatment. In the case of a GS 3+4=7 cancer being misdiagnosed as GS 3+3=6, the patient would receive a non-cancer diagnosis if GS 3+3=6 cancers are considered benign. This could have serious clinical implications and would have the effect of delaying or even preventing appropriate management.

CONCLUSION

Redesignating GS 3+3=6 prostatic adenocarcinoma as benign lacks sound scientific reasoning and is not the solution to prevent patients undergoing inappropriate definitive treatment. It would seem more appropriate to educate clinicians and patients regarding the indolent nature of this tumour and the appropriateness of active surveillance without radical treatment unless a higher grade component is identified.

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References

1. Gleason DF. Classification of prostatic carcinomas. *Cancer Chemother Rep* 1966; 50: 125–8.
2. World Health Organization classification of tumours. In: Humphrey PA, Moch H, Reuter VE, Ulbright TM, editors. *Pathology and Genetics of the Urinary System and Male Genital Organs*. Lyon: IARC Press, 2016.
3. Delahunt B, Miller RJ, Srigley JR, Evans AJ, Samaratunga H. Gleason grading: past, present and future. *Histopathology* 2012; 60: 75–86.
4. Eggener SE, Berlin A, Vickers AJ, Paner GP, Wolinsky H, Cooperberg MR. Low-grade prostate cancer: time to stop calling it cancer. *J Clin Oncol* 2022; 40: 3110–4.
5. Paner GP, Zhou M, Simko JP, Eggener SE, van der Kwast T. Renaming grade group 1 prostate "cancer" from a pathology perspective: a call for multidisciplinary discussion. *Adv Anat Pathol* 2023; Apr 17: https://doi.org/10.1097/PAP.0000000000000400.
6. Labbate CV, Paner GP, Eggener SE. Should grade group 1 (GG1) be called cancer? *World J Urol* 2022; 40: 15–9.
7. Iczkowski KA, Molina M, Egevad L, *et al*. Low-grade prostate cancer should still be labelled cancer. *BJU Int* 2022; 130: 741–3.
8. Carter HB, Partin AW, Walsh PC. Gleason score 6 adenocarcinoma: should it be labeled as cancer? *J Clin Oncol* 2012; 30: 4294–6.
9. Epstein JI. Is grade group 1 (Gleason score 3 + 3 = 6) adenocarcinoma of the prostate really cancer? *Curr Opin Urol* 2022; 32: 91–5.
10. Semsarian CR, Ma T, Nickel B, *et al*. Low-risk prostate lesions: an evidence review to inform discussion on losing the "cancer" label. *Prostate* 2023; 83: 498–515.
11. Kulac I, Haffner MC, Yegnasubramanian S, Epstein JI, De Marzo AM. Should Gleason 6 be labeled as cancer? *Curr Opin Urol* 2015; 25: 238–45.
12. Krajewska M, Krajewski S, Epstein JI, *et al*. Immunohistochemical analysis of bcl-2, bax, bcl-X, and mcl-1 expression in prostate cancers. *Am J Pathol* 1996; 148: 1567–76.
13. de Muga S, Hernández S, Agell L, *et al*. Molecular alterations of EGFR and PTEN in prostate cancer: association with high-grade and advanced-stage carcinomas. *Mod Pathol* 2010; 23: 703–12.
14. Netto GJ, Cheng L. Emerging critical role of molecular testing in diagnostic genitourinary pathology. *Arch Pathol Lab Med* 2012; 136: 372–90.
15. Reichard CA, Klein EA. Clinical and molecular rationale to retain the cancer descriptor for Gleason score 6 disease. *Nat Rev Urol* 2017; 14: 59–64.
16. Bell PD, Teramoto Y, Gurung PMS, Numbere N, Yang Z, Miyamoto H. The clinical significance of perineural invasion by prostate cancer on needle core biopsy: involvement of single versus multiple sextant sites. *Arch Pathol Lab Med* 2022; 146: 1252–7.
17. Anderson BB, Oberlin DT, Razmaria AA, *et al*. Extraprostatic extension is extremely rare for contemporary Gleason score 6 prostate cancer. *Eur Urol* 2017; 72: 455–60.
18. Hassan O, Han M, Zhou A, *et al*. Incidence of extraprostatic extension at radical prostatectomy with pure Gleason score 3+3=6 (grade group 1) cancer: implications for whether Gleason score 6 prostate cancer should be renamed "not cancer" and for selection criteria for active surveillance. *J Urol* 2018; 199: 1482–7.
19. Donin NM, Laze J, Zhou M, Ren Q, Lepor H. Gleason 6 prostate tumors diagnosed in the PSA era do not demonstrate the capacity for metastatic spread at the time of radical prostatectomy. *Urology* 2013; 82: 148–52.
20. Ross HM, Kryvenko ON, Cowan JE, Simko JP, Wheeler TM, Epstein JI. Do adenocarcinomas of the prostate with Gleason score (GS) ≤6 have the potential to metastasize to lymph nodes? *Am J Surg Pathol* 2012; 36: 1346–52.
21. Kweldam CF, Wildhagen MF, Bangma CH, van Leenders GJ. Disease-specific death and metastasis do not occur in patients with Gleason score ≤6 at radical prostatectomy. *BJU Int* 2015; 116: 230–5.
22. Delahunt B, Egevad L, Srigley JR, *et al*. Validation of International Society of Urological Pathology (IUSP) grading for prostatic adenocarcinoma in thin core biopsies using TROG 02.04 'RADAR' trial clinical data. *Pathology* 2015; 47: 520–5.
23. Liu JJ, Lichtensztajn DY, Gomez SL, *et al*. Nationwide prevalence of lymph node metastases in Gleason score 3+3=6 prostate cancer. *Pathology* 2014; 46: 306–10.
24. Eggener SE, Scardino PT, Walsh PC, *et al*. Predicting 15-year prostate cancer specific mortality after radical prostatectomy. *J Urol* 2011; 185: 869–75.
25. Hamdy FC, Donovan JL, Lane JA, *et al*. ProtecT Study Group. Fifteen-year outcomes after monitoring, surgery, or radiotherapy for prostate cancer. *N Engl J Med* 2023; 388: 1547–58.
26. Chen ME, Johnston D, Reyes AO, *et al*. A streamlined three-dimensional volume estimation method accurately classifies prostate tumors by volume. *Am J Surg Pathol* 2003; 27: 1291–301.

27. Samaratunga H, Delahunt B, Gianduzzo T, *et al*. The prognostic significance of the 2014 International Society of Urological Pathology (ISUP) grading system for prostate cancer. *Pathology* 2015; 47: 515–9.
28. Gandaglia G, Ploussard G, Valerio M, *et al*. The key combined value of multiparametric magnetic resonance imaging, and magnetic resonance imaging-targeted and concomitant systematic biopsies for the prediction of adverse pathological features in prostate cancer patients undergoing radical prostatectomy. *Eur Urol* 2020; 77: 733–41.
29. Mamawala MK, Meyer AR, Landis PK, *et al*. Utility of multiparametric magnetic resonance imaging in the risk stratification of men with grade group 1 prostate cancer on active surveillance. *BJU Int* 2020; 125: 861–6.
30. Samaratunga H, Delahunt B, Yaxley J, *et al*. Clinical significance of cancer in radical prostatectomy specimens: analysis from a contemporary series of 2900 men. *Pathology* 2014; 46: 11–4.
31. Alenda O, Ploussard G, Mouracade P, *et al*. Impact of the primary Gleason pattern on biochemical recurrence-free survival after radical prostatectomy: a single-center cohort of 1,248 patients with Gleason 7 tumors. *World J Urol* 2011; 29: 671–6.
32. Delahunt B, Steigler A, Atkinson C, *et al*. Percentage grade 4 tumour predicts outcome for prostate adenocarcinoma in needle biopsies from patients with advanced disease: 10-year data from the TROG 03.04 RADAR trial. *Pathology* 2022; 54: 49–54.
33. van Leenders GJLH, van der Kwast TH, Grignon DJ, *et al*. The 2019 International Society of Urological Pathology (ISUP) consensus Conference on grading of prostatic carcinoma. *Am J Surg Pathol* 2020; 44: e87–99.
34. Epstein JI, Amin MB, Fine SW, *et al*. The 2019 genitourinary Pathology Society (GUPS) white paper on contemporary grading of prostate cancer. *Arch Pathol Lab Med* 2021; 145: 461–93.
35. Amin MB, Lin DW, Gore JL, *et al*. The critical role of the pathologist in determining eligibility for active surveillance as a management option in patients with prostate cancer: consensus statement with recommendations supported by the College of American Pathologists, International Society of Urological Pathology, Association of Directors of Anatomic and Surgical Pathology, the New Zealand Society of Pathologists, and the Prostate Cancer Foundation. *Arch Pathol Lab Med* 2014; 138: 1387–405.
36. Epstein JI, Walsh PC, Carmichael M, Brendler CB. Pathologic and clinical findings to predict tumor extent of nonpalpable (stage T1c) prostate cancer. *JAMA* 1994; 271: 368–74.
37. Flach RN, Egevad L, Eklund M, *et al*. Use of the ISUP e-learning module improves interrater reliability in prostate cancer grading. *J Clin Pathol* 2022; Nov 3: jcp-2022-208506.
38. Egevad L, Ahmad AS, Algaba F, *et al*. Standardization of Gleason grading among 337 European pathologists. *Histopathology* 2013; 62: 247–56.
39. Egevad L, Delahunt B, Berney DM, *et al*. Utility of pathology imagebase for standardisation of prostate cancer grading. *Histopathology* 2018; 73: 8–18.